

Scalability & Future Plan

Date	04 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Credit Card Approval Prediction solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs on Flask's built-in development server (app.run(debug=True)), which is single-threaded and not built for concurrent traffic.	Cannot reliably handle multiple simultaneous prediction requests, risking slow responses or dropped requests under load.	High
2	The model is trained on a single static historical dataset (creditcard.csv) with no automated retraining pipeline.	Prediction accuracy may degrade over time as applicant profiles and credit patterns evolve.	Medium
3	No user authentication, request logging, or history tracking — only single, stateless predictions are supported.	No audit trail of past applications, making it difficult to track decisions or support compliance reviews.	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask development server handling one request thread at a time.	Deploy behind a production-grade WSGI server (Gunicorn or Waitress) with a load balancer and multiple worker processes to serve concurrent applicants.
2	Data Storage	Applicant dataset (creditcard.csv) and trained model (credit_card_model.pkl) stored as local flat files.	Migrate applicant records to a managed database (e.g. PostgreSQL) and store model artifacts in cloud object storage (e.g. AWS S3 / Azure Blob).
3	Performance	Model loaded once at startup; each prediction is processed synchronously per request.	Containerize the application with Docker, add response caching where applicable, and consider a dedicated model-serving layer (e.g. IBM Watson Machine Learning) for faster, scalable inference.

4	Security	No authentication; form inputs validated only with a basic try/except block.	Add HTTPS, stronger input validation/sanitization, role-based access control for analysts vs. customers, and secure storage/encryption of applicant PII.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add model explainability (e.g. SHAP / feature importance) to show why an application was approved or rejected.	1–2 months	Improves trust and transparency for analysts and customers reviewing decisions.
Phase 2	Introduce user authentication and application history tracking for analysts and customers.	2–3 months	Enables audit trails, personalized dashboards, and compliance-ready record keeping.
Phase 3	Move to production-grade deployment — containerize with Docker and serve via Unicorn/Waitress behind a reverse proxy, or host on a managed ML platform such as IBM Watson Machine Learning.	3–4 months	Delivers reliable, scalable real-time predictions capable of handling production traffic.
Phase 4	Build an automated pipeline to periodically retrain the model on new applicant data.	4–6 months	Keeps prediction accuracy high as applicant patterns and credit trends shift over time.